DETERMINATION OF CURRENT CIRCUIT USING MESH ANALYSIS

4/2/26 EXP NO.: 2

AIM:

To determine the current in circuit using mesh analysis both theoretically and practically for a given DC circuit.

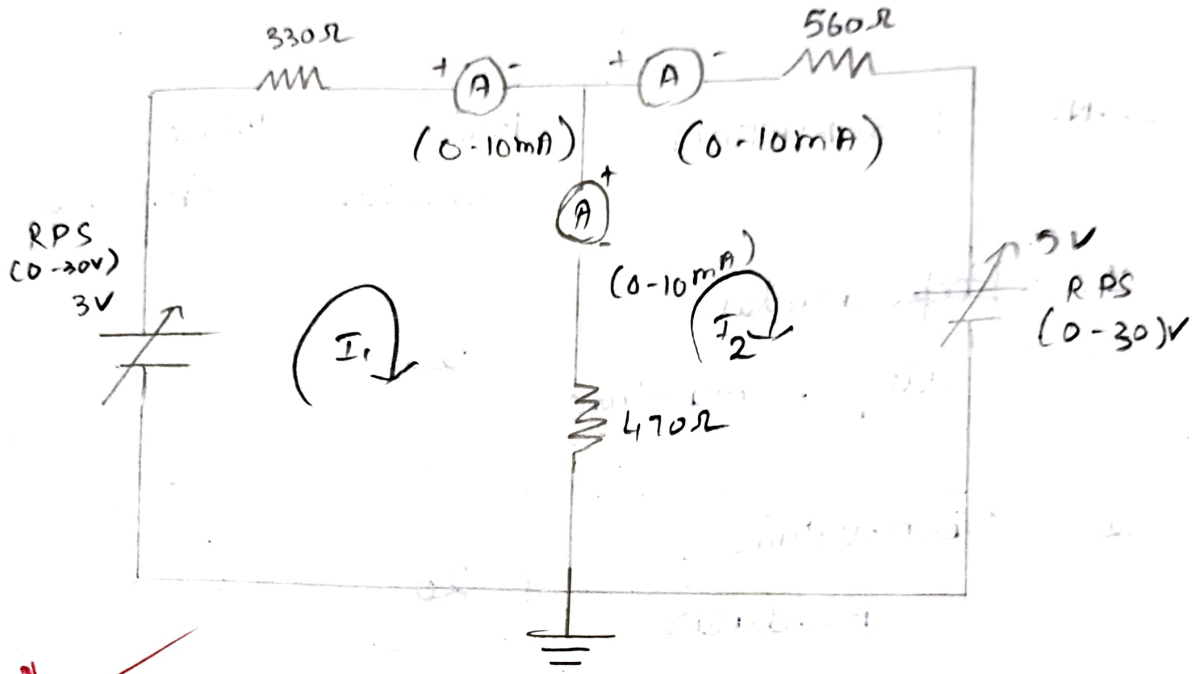
APPARATUS REQUIRED :

SL. NO.	APPARATUS	SPECIFICATION	QUANTITY
1.	Regulated Power supply	(0-30V)	1
2.	Ammeter	(0-10mA) MC	3
3.	Resistors	330Ω, 470Ω, 560Ω	Each 1
4.	Bread Board	—	1

PROCEDURE :

1. Give connections as per the circuit diagram.
2. Switch ON the supply, vary the RPS and set a particular input voltage.
3. Note down the readings and tabulate them.
4. Vary the RPS for different inputs and note down readings
5. Reduce the RPS to minimum value and switch off the supply
6. Using the tabulated values, verify the laws practically and theoretically

CIRCUIT DIAGRAM :



TABULAR COLOUMN :

Parameters	Theoretical	Practical
I_1	1.22 mA	1.4mA
I_2	-4.29 mA	-4.5mA
$I_1 - I_2$	5.51 mA	5.4 mA

o/p verified

CALCULATIONS :

$R_{11} = 800$ $\quad R_{21} = -470$

$R_{12} = -470$ $\quad R_{22} = 1030$

$$\Delta = \begin{vmatrix} 800 & -470 \\ -470 & 1030 \end{vmatrix} = 603100 \qquad \Delta_1 = \begin{vmatrix} 3 & -470 \\ -5 & 1030 \end{vmatrix} = 740$$

$$\Delta_2 = \begin{vmatrix} 800 & 3 \\ -470 & -5 \end{vmatrix} = 2590 \qquad I_1 = \Delta_1 / \Delta = \frac{740}{603100}$$

$$= 1.22 mA$$

$$I_2 = \Delta_2 / \Delta = \frac{2590}{603100} = -4.29 mA$$

$$I_1 - I_2 = 1.22 + 4.29 = 5.51 mA$$

MARKS SPLIT-UP :

SL. NO.	Description	Marks allotted	Marks Obtained
1.	Experimental setup / connections	20	20
2.	Observation / Readings	20	20
3.	Calculations / Results	20	20
4.	Viva	20	20
5.	Record	20	20

100

100

RESULT :

Thus, the Mesh analysis is verified practically and theoretically. The resultant currents for supply are :

a) The current across 330 Ω resistors is 1.4 mA

b) The current across 560 Ω resistors is -4.5 mA

c) The current across 470 Ω resistors is 5.4 mA.

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